4.6.1 Triple Integrals in Cylindrical Coordinates_Guide

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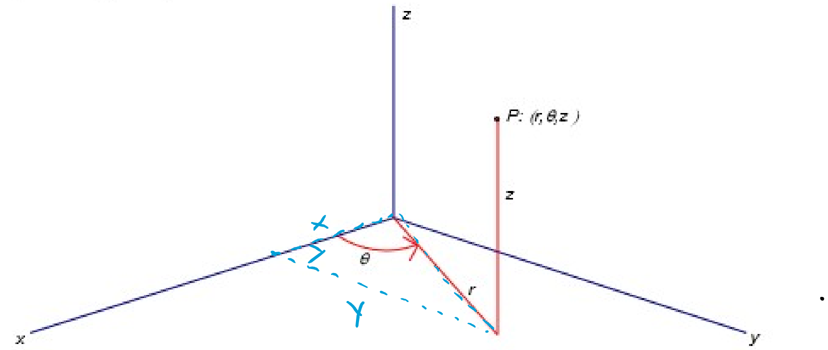
4.6.1 Triple Integrals in Cylindrical Coordinates

The cylindrical coordinate system is designed to help us find triple integrals over certain types of regions in three space. The cylindrical coordinate system is the three dimensional equivalent of polar coordinates.
A picture of a point plotted by its cylindrical coordinates is:



To Convert from Cylindrical to Rectangular Coordinates:

Use the following equations: $x = r\cos(\theta)$ $y = r\sin(\theta)$ $z = z$

To Convert from Rectangular Coordinates to Cylindrical Coordinates:

Use the following equations (assuming $x \neq 0$): $r^2 = x^2 + y^2$, $\tan(\theta) = \frac{y}{x}$ & $z = z$

Fact: Suppose that E is a region whose projection onto the xy-plane, D, is conveniently described in polar. Specifically, suppose that f is continuous and $E = \{(x,y,z) \mid (x,y) \in D, \underline{u_1(x,y)} \le z \le \underline{u_2(x,y)}\}$ where $D = \{(r,\theta) \mid \underline{\alpha} \le \theta \le \underline{\beta}, \underline{h_1(\theta)} \le r \le \underline{h_2(\theta)}\}$.

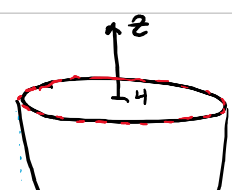
Then, by using what we know about double integrals in polar and triple integrals over type 1 regions, we have:

$$\iiint_E f(x,y,z)\,dV = \int_\alpha^\beta \int_{h_1}^{h_2} \int_{u_1}^{u_2} f(r\cos(\theta), r\sin(\theta), z)\, r\, dz\, dr\, d\theta$$

Note: One may modify the definition of cylindrical coordinates so that we view the projection onto the xz-plane or yz-plane as being described in polar. However, in this section we will not be considering such modifications. Therefore, when we set up a triple integral as an iterated integral in cylindrical we will always pick z as the innermost variable of integration and follow the general procedure from section 4.5 (keeping in mind that the projection of the solid onto the xy-plane must be described in polar).

$z = x^2 + y^2$ $z = 4$

Example 1: Evaluate $\iiint_E z\,dV$ where $E = \{(x,y,z) : z \ge x^2 + y^2,\ z \le 4\}$.



We see E has a clear "floor" & "ceiling".

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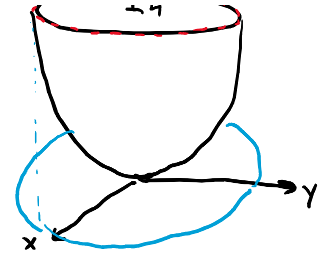


We see E has a clear "floor" & "ceiling".

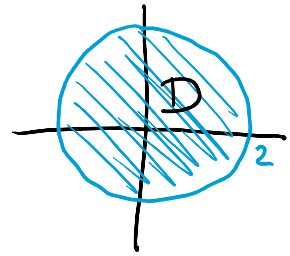
$r^2 = x^2+y^2 \le z \le 4$

We project E into the xy-plane.

We get a region D.

When $z=4$ we get $4=x^2+y^2$

$D=\{(r,\theta) \mid 0 \le r \le 2 \ \&\ 0 \le \theta \le 2\pi\}$

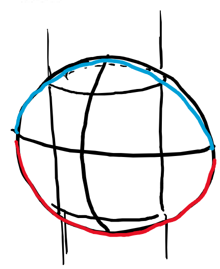


Therefore,

$$\iiint_E z\,dV = \int_0^{2\pi}\int_0^{2}\int_{r^2}^{4} z \cdot r\,dz\,dr\,d\theta$$

Example 2: Use cylindrical coordinates to set up (but do not evaluate) an iterated integral which yields the volume of the solid which lies within the cylinder $x^2+y^2=1$ and the sphere $x^2+y^2+z^2=4$.

We see the region is "pill shaped" & has a clear floor & ceiling (which are both the sphere)

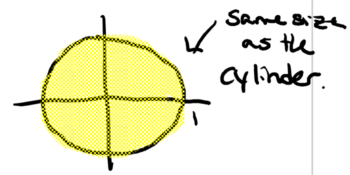


$x^2+y^2+z^2=4 \rightarrow z=\pm\sqrt{4-x^2-y^2}$

So $\underset{-\sqrt{4-r^2}}{-\sqrt{4-x^2-y^2}} \le z \le +\sqrt{4-x^2-y^2} = \sqrt{4-r^2}$

We then project into the xy-plane & get D

$D=\{(r,\theta) \mid 0 \le r \le 1 \ \&\ 0 \le \theta \le 2\pi\}$



So $V = \iiint_E 1\,dV = \int_0^{2\pi}\int_0^{1}\int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} 1\, r\,dz\,dr\,d\theta$

Example 3: Evaluate the following iterated integral by changing to cylindrical coordinates:

$$\int_{-3}^{3}\int_{0}^{\sqrt{9-x^2}}\int_{0}^{9-x^2-y^2}\sqrt{x^2+y^2}\,dz\,dy\,dx.$$

Note: Usually when students first see a problem like this, it is tempting for them to simply replace all the x

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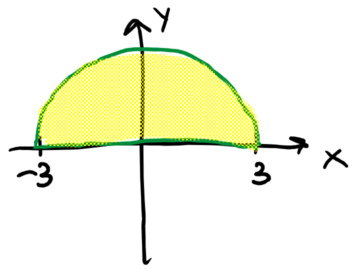
$$\int_{-3}^{3}\int_{0}^{\sqrt{9-x^2}}\int_{0}^{9-x^2-y^2}\sqrt{x^2+y^2}\,dz\,dy\,dx.$$

Note: Usually when students first see a problem like this, it is tempting for them to simply replace all the x variables with $r\cos(\theta)$ and all the y variables with $r\sin(\theta)$. As we will see this does not quite work in general.

We are integrating over $E = \{(x,y,z) \mid 0 \le z \le 9-x^2-y^2,\ 0 \le y \le \sqrt{9-x^2},\ \&\ -3 \le x \le 3\}$

The bounds on z can stay, but we need to redo the bounds on the projection D.

We sketch a picture of D.



Note $y=\sqrt{9-x^2} \rightarrow y^2 = 9-x^2 \rightarrow x^2+y^2=9$

Let $D = \{(r,\theta) \mid 0 \le r \le 3\ \&\ 0 \le \theta \le \pi\}$

So $$\int_{-3}^{3}\int_{0}^{\sqrt{9-x^2}}\int_{0}^{9-x^2-y^2}\sqrt{x^2+y^2}\,dz\,dy\,dx = \int_{0}^{\pi}\int_{0}^{3}\int_{0}^{9-r^2}\underbrace{\sqrt{r^2}}_{r^2}\,r\,dz\,dr\,d\theta$$